

Roozbeh Sarenche

Postdoctoral Researcher in Blockchain Security & Cryptography

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Education

PhD in Blockchain Security

KU Leuven, COSIC

Jul 2020 – Sep 2025

Leuven, Belgium

- **Thesis:** Incentive Analysis in Permissionless Blockchains (Advisors: Prof. Bart Preneel, Prof. Svetla Nikova)
- **Courses:** Cryptography and Network Security, Computer Algebra for Cryptography, Quantum Computing

MSc in Secure Communications and Cryptography

Sharif University of Technology

Sep 2016 – Dec 2018

Tehran, Iran

- **Thesis:** Holding Secure Auctions in Smart Grids
- **Selected Courses:** Cryptography Mathematics, Cryptography, Network Security, Advanced Cryptography

BSc in Electrical Engineering

University of Tehran

Sep 2012 – Sep 2016

Tehran, Iran

Professional Experience

Postdoctoral Researcher

KU Leuven, COSIC

Sep 2025 – Present

Leuven, Belgium

- Conducting advanced research on consensus mechanisms, blockchain incentive structures, and cryptographic protocols.

Remote Research Intern (Cryptography)

Nethermind

May 2024 – Jul 2024

Remote (UK-based)

- Collaborated with the zero-knowledge research team (under Dr. Albert Garreta) on designing a ZK proving system for verifiable machine-learning inference.
- Implemented and optimized advanced folding schemes, including Nova and HyperNova, to achieve efficient proof generation.

Publications

Journal Publications

1. **Selfish Mining Time-Averaged Analysis in Bitcoin: Is Orphan Reporting an Effective Countermeasure?**
R. Sarenche, R. Zhang, S. Nikova, B. Preneel. IEEE Transactions on Information Forensics and Security (TIFS), vol. 20, pp. 449–464, 2024.
Impact Factor: 8.0 | Citations: 15 | Lead Author
2. **DASLog: Decentralized Auditable Secure Logging for UAV Ecosystems**
R. Sarenche, F. Aghili, T. Yoshizawa, D. Singelée. IEEE Internet of Things Journal, vol. 10(23), pp. 20264–20284, 2023.
Impact Factor: 8.9 | Citations: 10 | Co-lead Author

3. **A Secure and Privacy-Preserving Protocol for Holding Double Auctions in Smart Grid**
R. Sarenche, M. Salmasizadeh, M.H. Ameri, M.R. Aref. *Information Sciences*, vol. 557, pp. 108–129, 2021.
Impact Factor: 6.8 | Citations: 42 | Lead Author

Conference Proceedings

4. **Bitcoin under Volatile Block Rewards: How Mempool Statistics Can Influence Bitcoin Mining**
R. Sarenche, A. Aghabagherloo, S. Nikova, B. Preneel. *ACM Conference on Computer and Communications Security (CCS 2025)*, pp. 903–917, 2025.
Acceptance Rate: 14% | Citations: 5 | Lead Author
5. **Commitment Attacks on Ethereum’s Reward Mechanism**
R. Sarenche, E. N. Tas, B. Monnot, C. Schwarz-Schilling, B. Preneel. *IEEE European Symposium on Security and Privacy (EuroS&P 2025)*, pp. 504–526, 2025.
Acceptance Rate: 20% | Citations: 3 | Co-lead Author
6. **Mining Power Destruction Attacks in the Presence of Petty-Compliant Mining Pools**
R. Sarenche, S. Nikova, B. Preneel. *Financial Cryptography and Data Security (FC 2025)*, pp. 208–224, 2025.
Acceptance Rate: 15% | Citations: 2 | Lead Author
7. **Deep Selfish Proposing in Longest-Chain Proof-of-Stake Protocols**
R. Sarenche, S. Nikova, B. Preneel. *Financial Cryptography and Data Security (FC 2024)*, LNCS 14744, pp. 24–40, Springer, 2024.
Acceptance Rate: 22% | Citations: 9 | Lead Author
8. **Fostering Energy Transition in Smart Cities: DLTs for Peer-to-Peer Electricity Trading**
M. Montakhabi, S. van der Graaf, A. Madhusudan, R. Sarenche, M.A. Mustafa. *17th International Conference on Distributed Computing in Sensor Systems*, 2021.
Citations: 13
9. **Distributed Ledger Technologies to Foster Energy Transition in Smart Cities: The Case of P2P Electricity Trading**
M. Montakhabi, S. van der Graaf, A. Madhusudan, R. Sarenche, M.A. Mustafa. *3rd IEEE International Workshop on Smart Circular Economy*, 2021.
Citations: 2

Preprints & ePrints

10. **Distributed Symmetric Key Establishment with Forward Secrecy for Implantable Medical Devices**
R. Sarenche, S. Duttagupta, F. Milizia, K. Bogner, V. Mishra. *Cryptology ePrint Archive*, 2025.
11. **Priv-PFL: A Privacy-Preserving and Efficient Personalized Federated Learning Approach**
A. Aghabagherloo, R. Sarenche, M. Zarezadeh, B. Preneel, S. Köpsell. *Cryptology ePrint Archive*, 2025.
Citations: 1

Selected Talks & Presentations

- **Cybersecurity Research Program Flanders (CIF) Workshop**: “Incentive Analysis in Permissionless Blockchains”, Oct 2025.
- **Cryptofinance Seminar (Bitcoin.fr)**: “Time Averaged Analysis of Selfish Mining in Bitcoin”, Nov 2024.
- **Iranian Society of Cryptology Seminar**: “Security Analysis of Longest-Chain-Based Consensus Mechanisms”, Nov 2024.
- **Accepted Speaker, Ethereum Foundation Devcon**: “A DAG-Based Mechanism for Fairer and More Decentralized Reward Distribution”, Nov 2024.

Supervision & Mentoring

Master Student Supervision

- **Prabal Das**, “Ethereum Future Roadmap: Encrypted Mempools,” 2025
- **Timo Van Eetvelde**, “Blockchain Mining Games with Multiple Mining Pools,” 2024
- **Sayanil Ghosh**, “Increasing Bitcoin Revenue by Leveraging Rational Miners,” 2024
- **Shubh Goel**, “Using AI to Attack the Blockchain,” 2024
- **Axelle Moortgat**, “On-Chain Storage of NFT Collections on the Ethereum Network,” 2023

Bachelor Student Supervision

- **Joshua Frias**, “Blockchain Application for Digital Monitoring, Reporting, and Verification of CO2 Emissions,” 2025

Internship Supervision

- **Francesco Milizia**, “Blockchain-Based Authentication Protocol with Perfect Forward Secrecy,” Summer 2025. *This internship is part of a broader project involving five PhD students at COSIC, coordinated by me.*

Academic Service

- **Peer Review:** Reviewer for IEEE TIFS, Journal of Supercomputing, IEEE TAI, Digital Finance, PNAS Nexus, and sub-reviewer for FC, Eurocrypt, and Asiacrypt (2021–2025).
- **Event Organization:** Co-organizer of the Fifth ISC Winter School on Information Security and Cryptology (IoT Security focus, Feb 2025).

Open-Source Implementations

- **Smart-Contract Auditing System:** Implemented a smart-contract-based secure logging system (DASLog) on Hyperledger Besu.
- **Reinforcement Learning Models:** Implemented Deep Q-Learning and A3C reinforcement learning algorithms for analyzing complex blockchain proposing strategies.
- **Markov Decision Processes:** Developed MDP tools to computationally analyze the time-averaged profitability of selfish mining and related incentive attacks.
- **Consensus Deployment:** Implemented and deployed a novel blockchain consensus mechanism (DAG-Vote LMD GHOST) across distributed AWS servers.